

Functional Requirements

The **Optimizing User, Group, and Role Management with Access Control and Workflows** project is developed to provide a secure and efficient project management environment in ServiceNow. The following functional requirements define the core features and system capabilities required to achieve the project's objectives.

1. User Management

The system shall support user management by creating and maintaining user accounts for the project participants.

Requirements:

- Create two users: **Alice** (Project Manager) and **Bob** (Team Member).
- Assign appropriate roles and responsibilities based on each user's job function.
- Allow authenticated users to access the application according to their assigned permissions.
- Maintain user information required for project management activities.

2. Group Management

The system shall organize users into groups to simplify task assignment and access management.

Requirements:

- Create a **Project Team** group.
- Add Alice and Bob as members of the Project Team group.
- Enable group-based task assignment and collaboration.
- Allow the Project Manager to manage team-related activities efficiently.

3. Role Definition

The system shall implement Role-Based Access Control (RBAC) by defining specific roles for users.

Project Manager Role

The Project Manager shall have permission to:

- Create and manage projects.
- Create, assign, update, and delete project tasks.
- View all project and task records.

- Monitor project progress.
- Approve completed tasks.
- Close projects after successful completion.

Team Member Role

The Team Member shall have permission to:

- View assigned tasks.
- Update task status.
- Add work notes and comments.
- Mark assigned tasks as completed.

The Team Member shall not have permission to:

- Create projects.
- Modify project details.
- Delete project records.
- View or edit unauthorized project information.

4. Table Configuration

The system shall include custom tables to store project and task information.

Project Table

The Project table shall contain information related to each project, including:

- Project ID
- Project Name
- Description
- Project Manager
- Start Date
- End Date
- Status
- Priority

Task Table

The Task table shall store project task information, including:

- Task Number
- Task Name
- Project Reference
- Assigned To
- Assignment Group
- Status
- Priority
- Due Date
- Comments
- Work Notes

These tables shall maintain relationships between projects and their associated tasks.

5. Application Access

The application shall provide controlled access to modules based on user roles.

Requirements:

- Configure application modules for the Project and Task tables.
- Ensure only authorized users can access application modules.
- Allow Project Managers to access all project management modules.
- Allow Team Members to access only task-related modules assigned to them.
- Hide unauthorized modules from users without the required roles.

6. Access Control Lists (ACLs)

The system shall enforce security using Access Control Lists (ACLs) to protect project information.

Project Manager Permissions

The Project Manager shall have:

- Create access
- Read access
- Update access
- Delete access

- Full administrative control over project and task records

Team Member Permissions

The Team Member shall:

- Read only assigned tasks.
- Update only the **Status**, **Comments**, and **Work Notes** fields of assigned tasks.
- Be restricted from editing project information or deleting records.
- Be prevented from accessing records that are not assigned to them.

These ACLs ensure secure access while maintaining data integrity and accountability.

7. Workflow Automation (Flow Designer)

The system shall automate project management activities using ServiceNow Flow Designer.

Workflow Process

1. The Project Manager creates a new project.
2. Project tasks are created and assigned to the Team Member.
3. The Team Member receives an automatic notification for the assigned task.
4. The Team Member updates the task status and work notes as progress is made.
5. When the task is marked as completed, the system automatically notifies the Project Manager.
6. The Project Manager reviews the completed task and approves it if all requirements are satisfied.
7. The project status is updated based on task completion, ensuring accurate progress tracking.

Workflow automation reduces manual effort, improves communication, accelerates task completion, and enhances overall operational efficiency.